

SMART AMBULANCE BOOKING AND EMERGENCY RESPONSE SYSTEM

FULL-STACK WEB DEVELOPMENT

SUBMITTED BY

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Task 1: Product Requirements Document (PRD)

1. What is a PRD?

A Product Requirements Document (PRD) describes what a product should do, who it is for, and what features it should provide. It helps the development team understand the project's goals, user requirements, and expected functionality before development begins.

2. Project Goals and Objectives

The main goal of the Smart Ambulance Booking and Emergency Response System is to provide a fast, reliable, and organized platform for requesting ambulances and coordinating emergency response.

Objectives:

- Allow users to book an ambulance quickly.
- Show available ambulances based on location and status.
- Share the patient's pickup location.
- Provide live ambulance tracking and estimated arrival time.
- Connect patients with ambulance drivers and emergency services.
- Reduce delays in emergency response.
- Provide an organized booking and emergency history.
- Improve communication during emergencies.

3. Features and Requirements

User Features:

- User registration and login.
- Search and request available ambulances.
- Select ambulance type.
- Enter pickup and destination details.
- Share live location.
- Track ambulance in real time.
- Receive booking and arrival notifications.
- View booking and emergency history.

Task 1: Product Requirements Document (PRD)

Functional Requirements:

- 1. The system should allow users to create an account and log in securely.
- 2. Users should be able to search for nearby available ambulances.
- 3. Users should be able to enter pickup and destination details.
- 4. The system should create and confirm ambulance bookings.
- 5. The system should assign an available ambulance and driver.
- 6. The system should share the patient's location with the assigned driver.
- 7. The system should display the ambulance's live location and estimated arrival time.
- 8. Users should receive booking, dispatch, and arrival notifications.
- 9. Emergency requests should be prioritized for faster response.
- 10. Administrators should manage ambulances, drivers, bookings, and emergency records.

Non-Functional Requirements:

- Simple and user-friendly interface.
- Fast response time.
- Reliable service availability.
- Secure personal and location information.
- Accurate location tracking.
- Scalable system.
- Mobile-friendly interface.
- Maintainable and well-documented code.

4. How the PRD Helps

- Clearly defines what the application needs to provide.
- Identifies user requirements before development.
- Prevents unnecessary features and scope changes.
- Provides a clear development roadmap.
- Improves communication between developers and stakeholders.
- Helps developers and testers understand expected functionality.
- Reduces errors and rework.

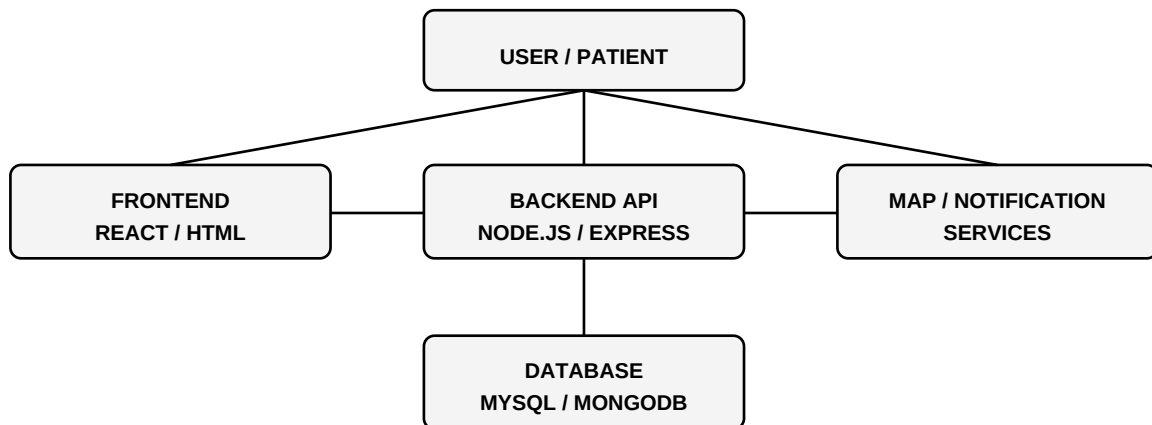
Task 2: Technical Requirements Document (TRD)

1. What is a TRD?

A Technical Requirements Document (TRD) explains how the project will be technically developed. It describes the system architecture, technologies, database, development workflow, APIs, and technical requirements needed to implement the system.

2. Technical Architecture

The application follows a layered full-stack architecture in which the frontend communicates with backend APIs, the backend manages business logic and databases, and external services provide maps, location, and notifications.



3. Main Technical Components

Frontend:

- Responsive user interface and dashboard.
- Registration and login pages.
- Ambulance search and booking pages.
- Live tracking map.
- Booking status and notification pages.

Backend:

- Authentication and authorization.
- Ambulance and driver management.
- Booking and emergency processing.
- Location and tracking APIs.
- Notification services.

Task 2: Technical Requirements Document (TRD)

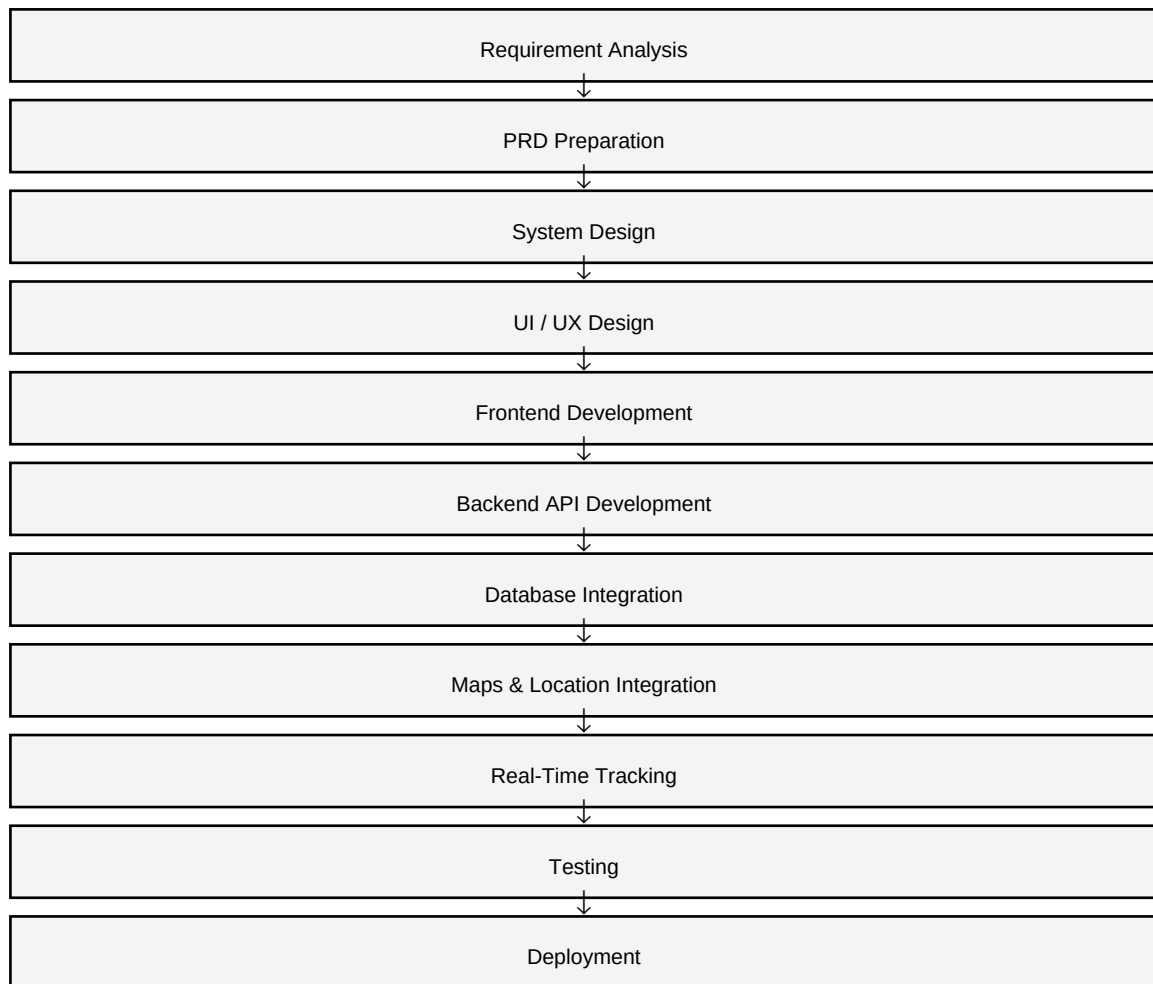
Database stores:

- User profiles and roles.
- Ambulance and driver information.
- Ambulance availability and status.
- Booking and emergency request details.
- Pickup and destination locations.
- Live location coordinates.
- Notifications and booking history.
- System logs.

4. Technologies and Tools

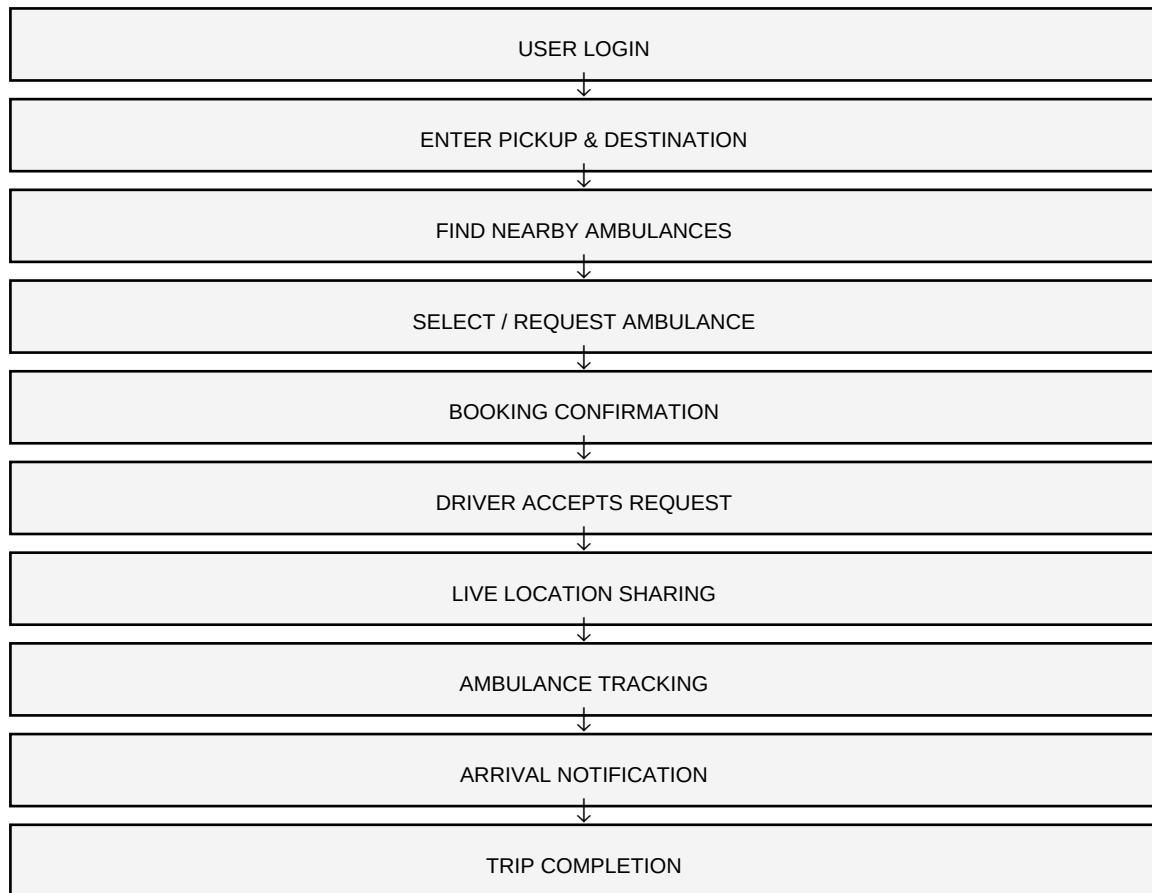
Category	Technology
Frontend	HTML, CSS, JavaScript / React.js
Backend	Node.js / Express.js
Database	MySQL / MongoDB
Maps & Location	Google Maps API / Mapbox
Real-Time Tracking	WebSocket / Socket.IO
Authentication	JWT / Firebase Authentication
Notifications	Firebase Cloud Messaging / Email API
Version Control	Git & GitHub
Development	VS Code
Testing	Browser / Postman
Deployment	AWS / Render / Vercel

5. Development Workflow



Task 2: Technical Requirements Document (TRD)

6. Ambulance Booking Workflow



7. Security Requirements

- Secure authentication and role-based access.
- JWT or session-based authorization.
- Input validation and secure API endpoints.
- Encrypted sensitive information.
- Protection of user phone and location data.
- Secure database access and backups.

8. Testing & Deployment

- Unit testing for individual modules.
- Integration testing for frontend and backend.
- API testing using Postman.
- Database testing.
- Location and real-time tracking testing.
- Performance and security testing.
- Cloud deployment with Git/GitHub.

9. How the TRD Helps

- Clearly explains the technical requirements.
- Defines system architecture and component interaction.
- Identifies the required technologies and APIs.

- Makes development organized and easier to execute.
- Helps developers understand database and backend requirements.
- Makes testing and debugging easier.
- Improves maintainability and scalability.
- Provides technical documentation for future development.

Task 3: Abstract

Project Title

“Smart Ambulance Booking and Emergency Response System”

Abstract

The Smart Ambulance Booking and Emergency Response System is a full-stack web application designed to make ambulance booking and emergency coordination faster, easier, and more reliable. In emergency situations, delays in finding an available ambulance, sharing the correct location, and communicating with drivers can increase response time.

The proposed system provides a centralized platform where users can register, request nearby ambulances, enter pickup and destination details, share their live location, and track the assigned ambulance. Ambulance drivers and administrators can manage requests, update availability, and coordinate emergency trips.

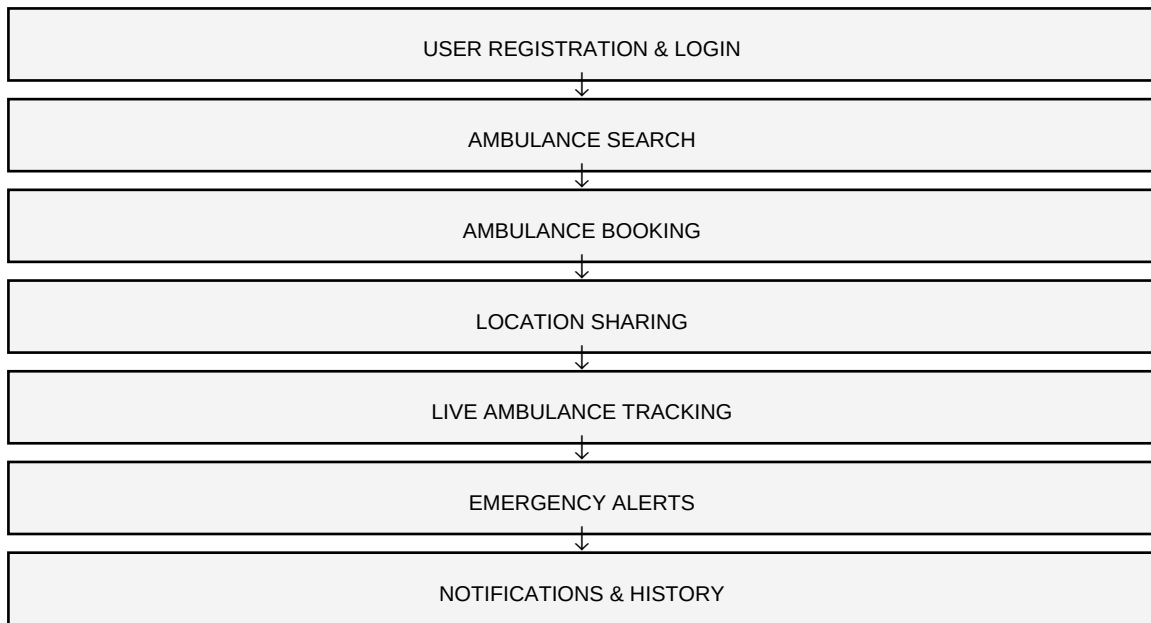
The system integrates web technologies, backend APIs, databases, maps and location services, real-time communication, and notification services. It maintains booking and emergency records so that users and administrators can review previous requests.

The expected outcomes are faster ambulance discovery, reduced response and communication delays, improved location accuracy, better ambulance utilization, and a more organized emergency response process.

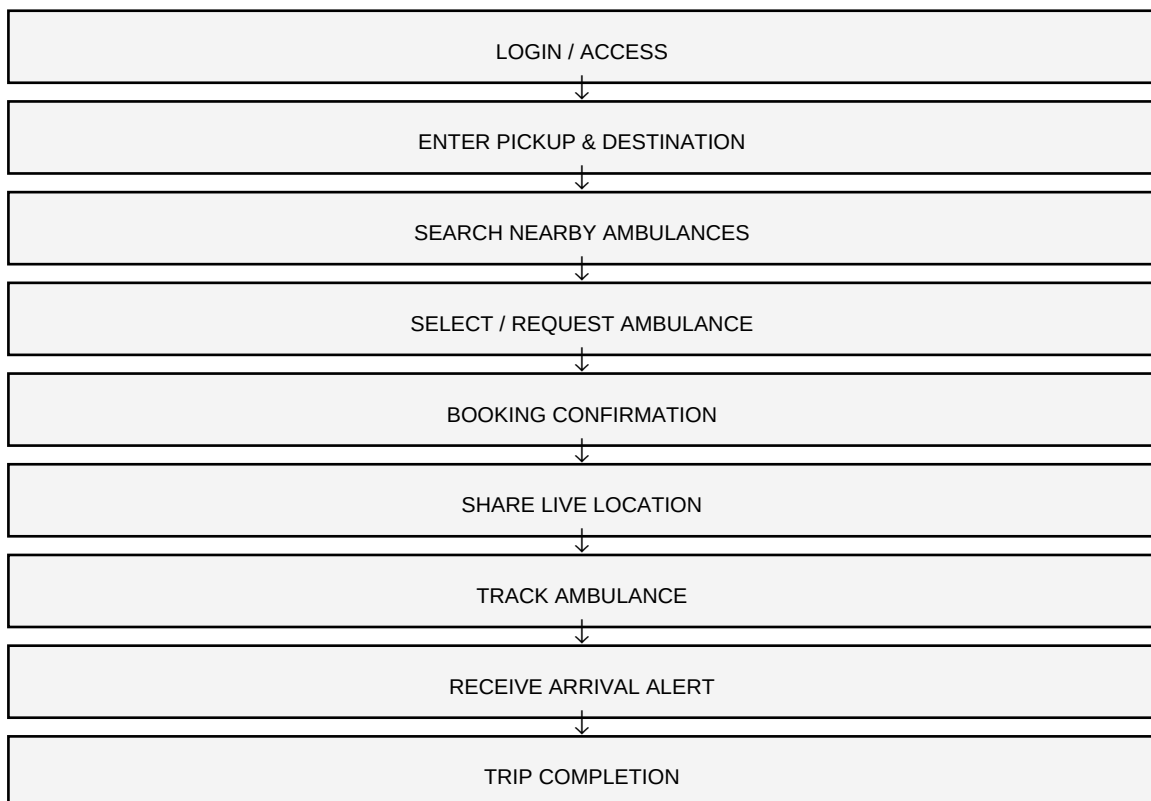
Overall, the project solves the limitations of manual ambulance searching and communication by providing a centralized, real-time, user-friendly digital platform for ambulance booking, tracking, and emergency coordination.

Task 4: Visual Representations

PRD Visual — Feature Overview

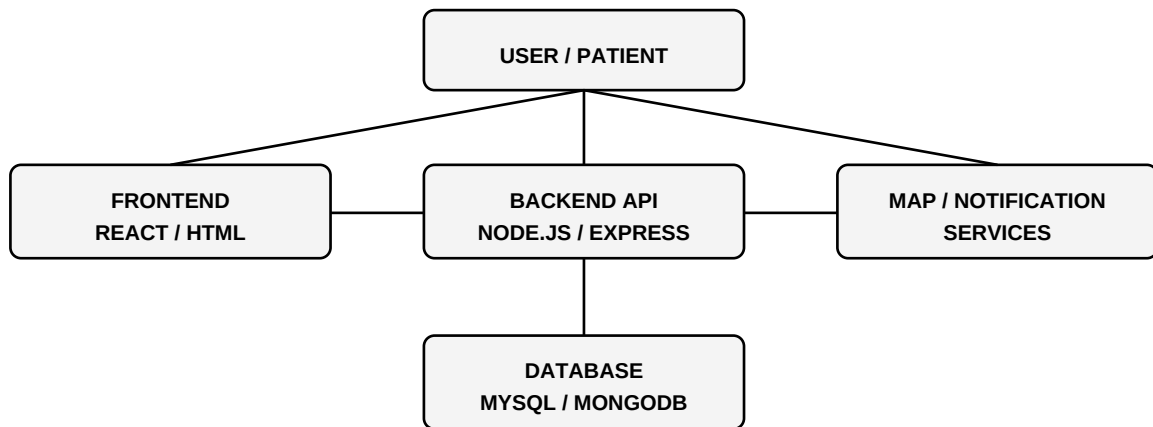


User Journey Diagram



Task 4: Visual Representations

TRD Visual — System Architecture



Functional Requirements Diagram

USER → LOGIN → SEARCH AMBULANCE → BOOK → SHARE LOCATION → DRIVER ACCEPTS → TRACK → ARRIVAL → COMPLETE

Feature Overview

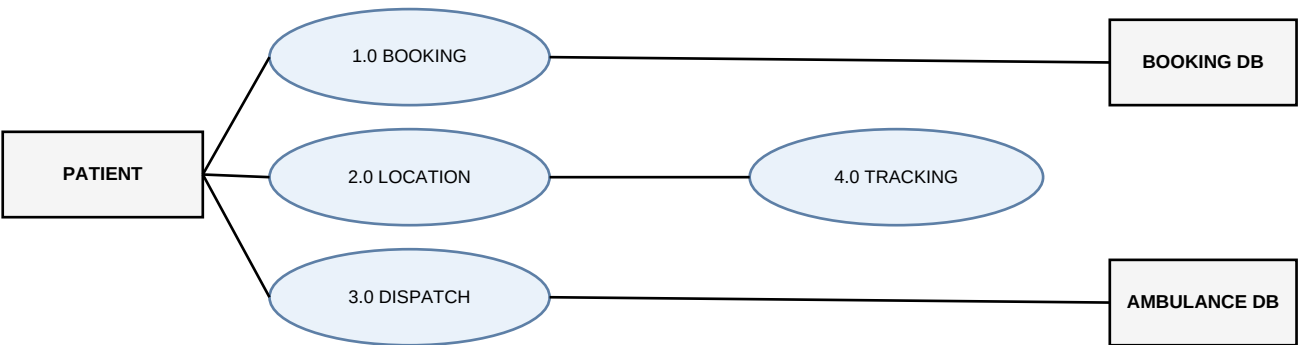
Ambulance Booking Search Request & confirmation	Location Services Pickup location Live GPS	Real-Time Tracking Driver location ETA
Emergency Response Priority requests Alerts	Driver Management Availability Trip updates	Admin & Reports Bookings History & analytics

System Diagrams

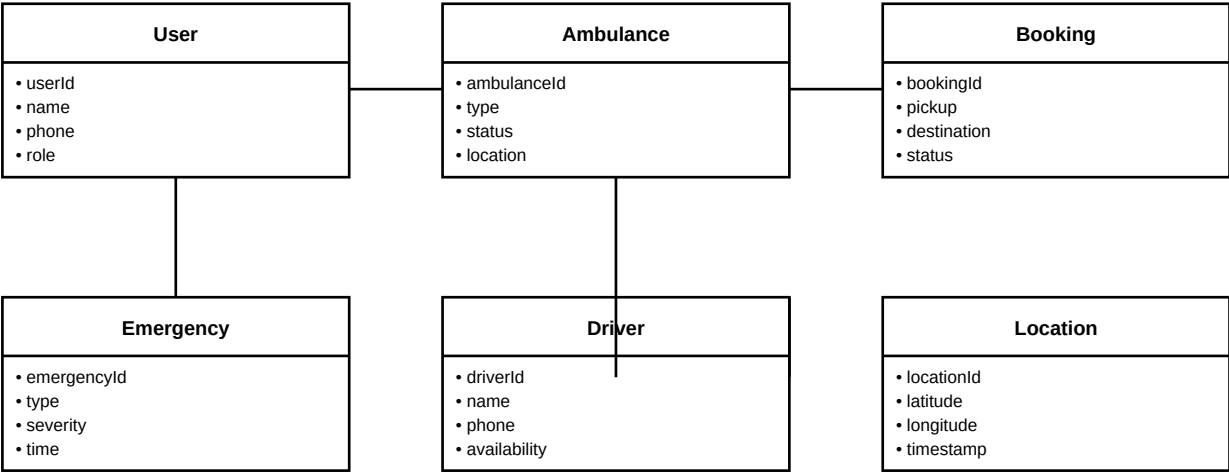
Use Case Diagram



Data Flow Diagram



Class Diagram



Main Classes

Class	Main Attributes / Responsibility
User	userId, name, phone, email, role
Ambulance	ambulanceId, type, status, location
Booking	bookingId, pickup, destination, status
Emergency	emergencyId, type, severity, time
Driver	driverId, name, phone, availability
Location	locationId, latitude, longitude, timestamp

Conclusion: The Smart Ambulance Booking and Emergency Response System combines full-stack web development, real-time location services, database management, APIs, and notifications to provide a faster and more organized ambulance booking and emergency response experience.